Minimum Spanning Trees:

Given a connected, undirected graph $G=(V,E)$ where V is the set of vertices, E is the set of possible ~~interconn~~ edges between each pairs of vertices, and for each edge $(u,v) \in E$, a weight $w(u,v)$ is associated which specifies the cost to connect u and v.

22 Friday

Goal: To find an acyclic subset $T \subseteq E$ that connects all of the vertices and whose total weight

$$w(T) = \sum_{(u,v) \in T} w(u,v)$$

is minimized.

$\because$ T is acyclic and connects all of the vertices, it must form tree and is termed as a spanning tree since it "spans" the graph G.

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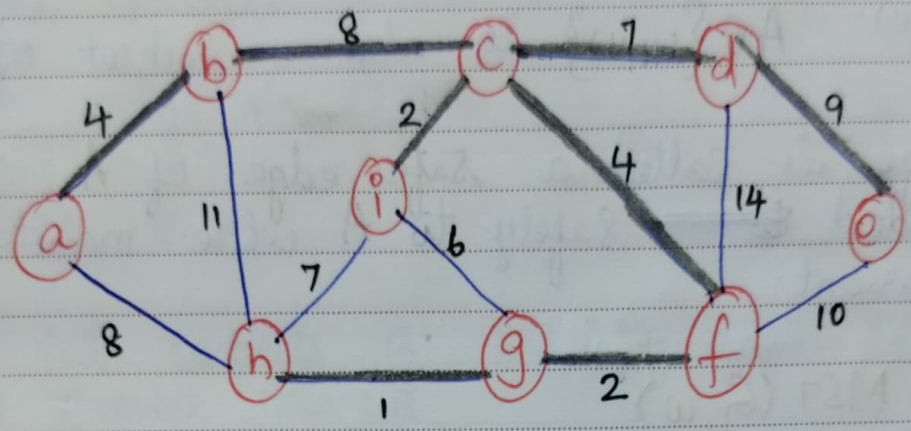
Saturday 23

The prob of determining the tree T is called minimum-spanning-tree problem.



Total weight - 37.

MSTs: a-b, b-c, c-d, c-f, d-e, c-i, h-g, g-f
a-b, a-h, c-d, c-f, d-e, c-i, h-g, g-f.

Sunday 24

Growing a MST:

Assume a connected, undirected graph $G=(V,E)$ with a weight fn $w: E \to R$, the goal is to find a <u>MST</u> for G.

* The greedy strategy grows the MST one edge at a time.

The generic method manages a set of edges A, maintaining the following loop invariant:

→ Prior to each iteration, A is a subset of some MST

→ At each step, determine an edge (u,v)

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that can be added to A without violating this invariant

(ie) $A \cup \{(u,v)\}$ is also a subset of MST.

This edge is called a safe edge of A, ∵ it can be added safely to A while maintaining the invariant.

Generic-MST (G, w)

```
1  A = ϕ
2  while A does not form a spanning tree
3      find an edge (u,v) that is safe for A
4      A = A ∪ {(u,v)}
5  return A.
```

26 Tuesday

The loop invariant is used as follows:

Initialization: After line 1, the set A trivially satisfies the loop invariant.

Maintenance: The while loop maintains the invariant by adding only safe edges.

Termination: All edges added to A are in a MST, and so the set A returned must be a MST.

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Wednesday 27

Defns:

1. Cut:

A cut $(S, V-S)$ of an undirected graph $G=(V,E)$ is a partition of V.

An edge $(u,v) \in E$ <u>crosses</u> the cut $(S, V-S)$ if one of its endpoints is in S and the other is in $V-S$.

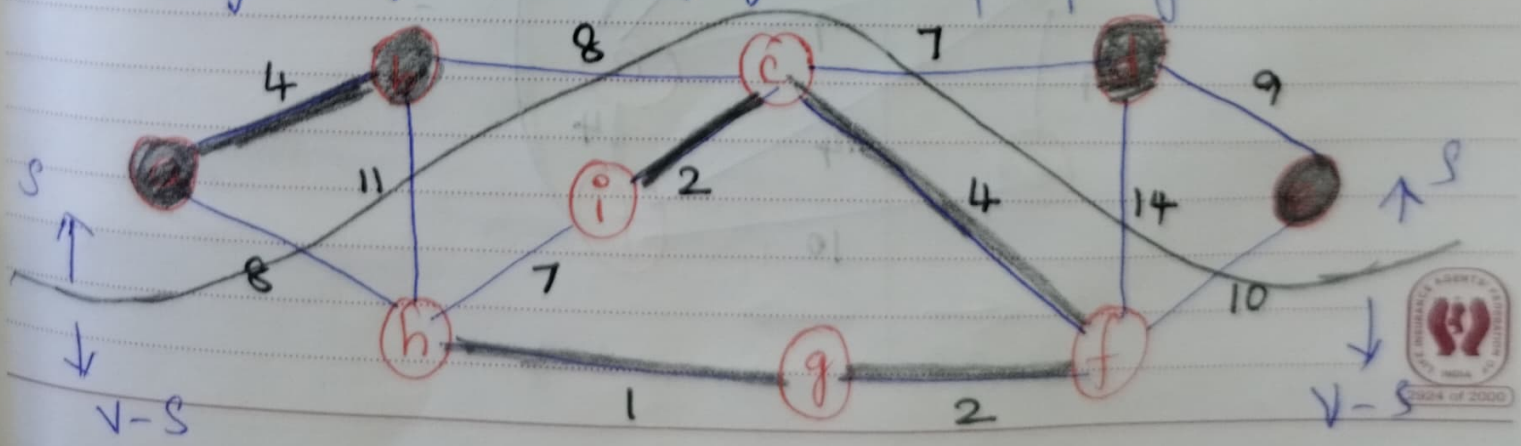
A cut <u>respects</u> a set A of edges if no edge in A crosses the cut.

An edge is a <u>light edge</u> crossing a cut if its weight is the minimum of any edge crossing the cut.

Thursday 28

Note: There can be more than one light edge crossing a cut in the case of ties.

In general, An edge is a <u>light edge</u> satisfying a given property if its weight is the minimum of any edge satisfying the property.



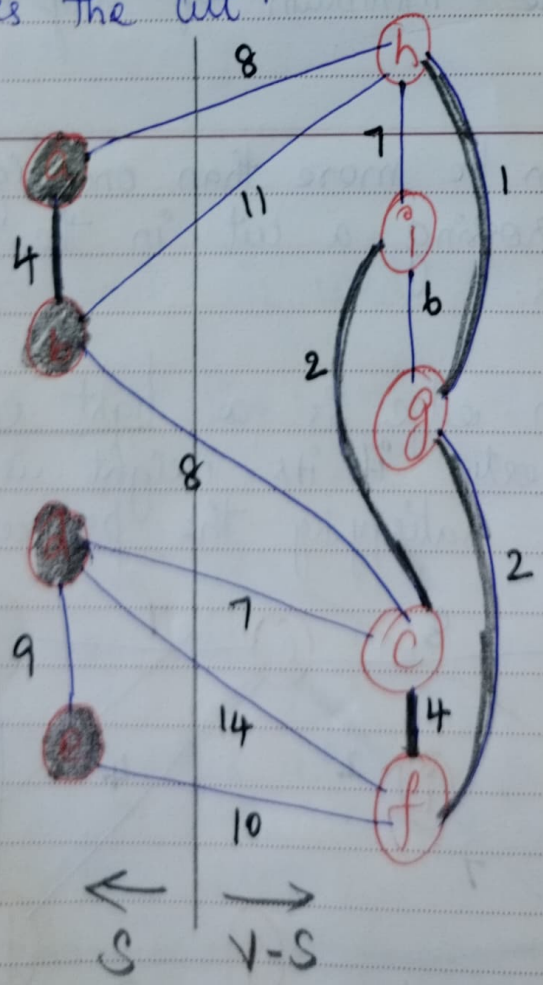
29 Friday

- Black vertices are in the set S.
  & white " " " " " V-S

- The edges crossing the cut are those connecting white vertices with black vertices.

- The edge (d,c) is the unique light edge crossing the cut.

- A subset A of the edges is shaded. The cut (S, V-S) respects A, ∵ no edge of A crosses the cut.



30 Saturday

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- The same graph with the vertices in the S on the left and the vertices in the set V-S on the right

An edge crosses the cut if it connects a vertex on the <u>left</u> with a vertex on the <u>right</u>.

Theorem:

Let $G=(V,E)$ be a connected, undirected graph with a real-valued weight fn w defined on E. Let A be a subset of E that is included in some minimum spanning tree for G, let $(S, V-S)$ be any cut of G that respects A, and let (u,v) be a light edge crossing $(S, V-S)$. Then, edge (u,v) is safe for A.